# Development of the University of Canterbury Four-Alternative Forced Choice (UC4AFC) Monosyllabic Word List

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## Introduction

- Auditory processing disorder (APD) may present as listening difficulties despite normal peripheral hearing
- The UCAST-FW (O'Beirne et al., 2012; Rickard et al., 2013) is an adaptive low-pass-filtered speech test used in APD assessment
- Original UCAST-FW stimuli (NU-CHIPS) are less suitable for NZ/Australian children
- To address this, the UC4AFC monosyllabic word list was developed using NZ/Australian English phonemic contrasts
- Initial 98-word list (created by Murray and O'Beirne; Murray, 2012) included:
  - target word
  - two phonemic foils
  - one semantic or unrelated foil
- Here we summarise three refinement and validation studies supporting paediatric clinical use

## Study 1 – Recording words, selecting images (Anderson, 2024)

**Objectives:**

i) Select representative images for each of the 98 target words
ii) Create audiovisual recordings for future AV-integration studies
iii) Evaluate low-pass-filter intelligibility characteristics of the recordings

**Method:**

- 392 candidate images collected (4 per target word)
- 43 adults selected "best image for children <12 years"
- Audiovisual recordings created for all target words
- 35 adults completed auditory-alone identification under 8–9 low-pass filter conditions (200–1500 Hz)
- Responses collected in:
  - open-set format
  - closed-set four-alternative forced choice (4AFC) format
- Data used to evaluate low-pass-filter psychometric characteristics for word list normalisation *(data not shown)*

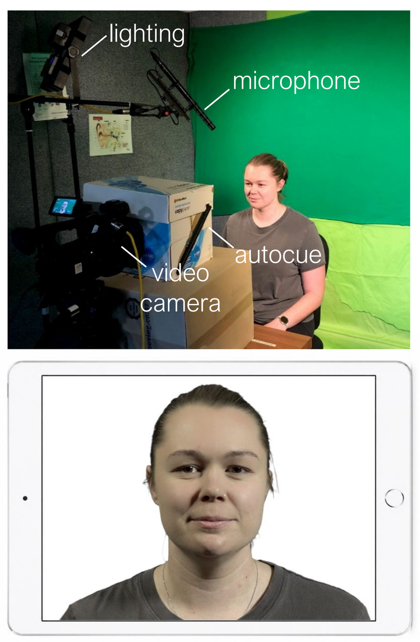



**Figure 1:** Audiovisual recording process. Setup (top) and post-processed frame.

**Results:**

- Strong image agreement was observed for several target words, including: chalk (98%), comb (95%), cap (86%), dirt (81%), leg (81%).
- Low-pass-filter results informed later item selection

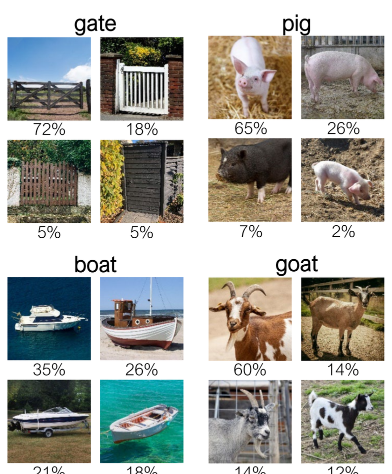



**Figure 2** *(left)*: Example vote distributions for candidate images corresponding to the target words *gate*, *pig*, *boat*, and *goat*.

**Figure 3** *(right)*: A finalised UC4AFC test plate for the target word *goat* with foils *boat*, *gate*, and *pig*.

## Study 2 – Spontaneous naming by 4-year-olds (Chong, 2025)

**Objectives:**

i) Evaluate whether four-year-olds consistently produced intended labels
ii) Quantify naming variability and naming latency

**Method:**

- 30 four-year-old children from low socioeconomic backgrounds participated
- Children viewed:
  - 98 selected UC4AFC images, 17 alternative candidate images
- Images presented with prompting phrases ("This is…", "These are…", etc.)
- Naming latency measured using synchronised audiovisual recordings
- Analyses included:
  - intended name agreement (the response matched our intended name)
  - target word inclusion (the target word was included within a longer response)
  - naming diversity using entropy statistic H (Snodgrass & Vanderwart, 1980) expressed as a percentage of the theoretical maximum for the sample size

**Results:**

Naming diversity ranged from near 0% (high agreement/low variability) to ~90% of theoretical maximum entropy (low agreement/high variability).

**Figure 4:** Intended name agreement and naming diversity

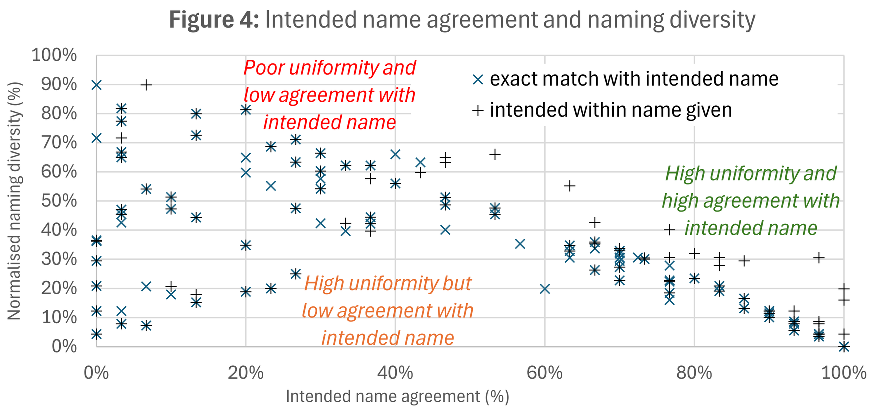
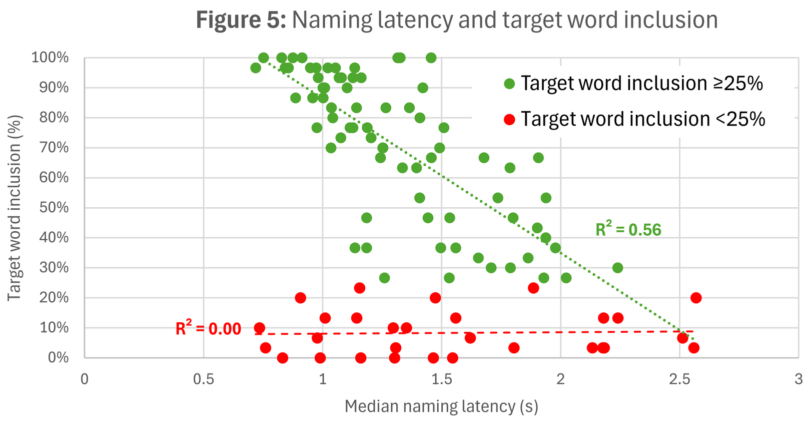



**Figure 5:** Naming latency and target word inclusion




## Study 3 – Receptive identification by 4-year-olds (Riley, 2026)

**Objective:** Determine whether four-year-olds could accurately and rapidly identify intended targets in 4AFC format

**Method:**

- 45 four-year-old children completed receptive 4AFC task
- Accuracy and response latency recorded for all 98 target items

**Results:**

- Approximately three quarters of items demonstrated:
  - >75% identification accuracy
  - median latency <3.5 s
- These items identified as strong candidates for final inclusion
- Poorer-performing items considered for exclusion

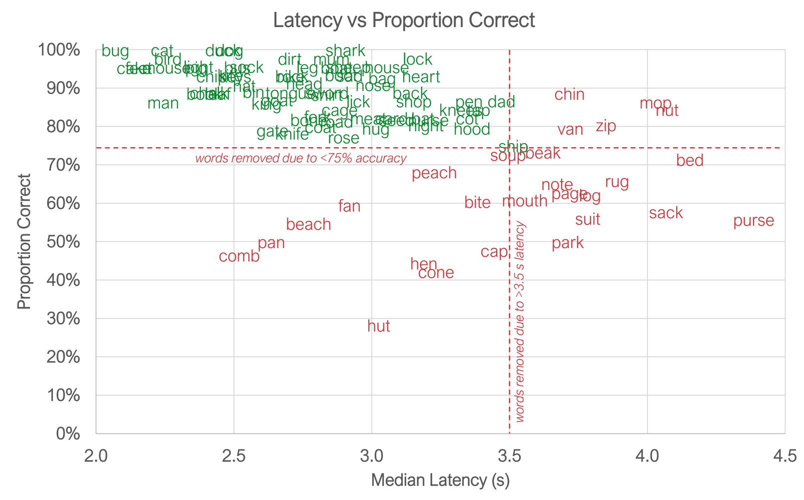



**Figure 6:** Median receptive identification latency versus proportion correctly identified for each target item.

## Final list composition

Final inclusion decisions incorporated:

- low-pass filter normalisation results
- expressive naming results
- receptive identification results
- phonemic foil balance requirements

**Table 1:** Finalised UC4AFC word lists and foil categories

| # | Target | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | lock | sock (V, FINAL C) | lick (INITIAL C, FINAL C) | knees (UNRELATED) |
| 2 | kite | night (V, FINAL C) | cot (INITIAL C, FINAL C) | chalk (SEMANTIC) |
| 3 | coat | boat (V, FINAL C) | kite (INITIAL C, FINAL C) | sock (SEMANTIC) |
| 4 | bat | hat (V, FINAL C) | back (INITIAL C, V) | duck (SEMANTIC) |
| 5 | keys | knees (V, FINAL C) | sheep (V) | house (SEMANTIC) |
| 6 | feet | meat (V, FINAL C) | seed (V) | nose (SEMANTIC) |
| 7 | shirt | dirt (V, FINAL C) | shark (INITIAL C) | bin (UNRELATED) |
| 8 | dirt | shirt (V, FINAL C) | duck (INITIAL C) | card (UNRELATED) |
| 9 | fork | chalk (V, FINAL C) | feet (INITIAL C) | nose (UNRELATED) |
| 10 | sad | dad (V, FINAL C) | sock (INITIAL C) | light (UNRELATED) |
| 11 | bin | chin (V, FINAL C) | bug (INITIAL C) | shark (UNRELATED) |
| 12 | house | mouse (V, FINAL C) | head (INITIAL C) | gate (SEMANTIC) |
| 13 | sock | rock (V, FINAL C) | seed (INITIAL C) | boot (SEMANTIC) |
| 14 | boat | coat (V, FINAL C) | bike (INITIAL C) | shark (SEMANTIC) |
| 15 | chip | chin (INITIAL C, V) | chalk (INITIAL C) | nurse (UNRELATED) |
| 16 | night | knife (INITIAL C, V) | coat (FINAL C) | leaf (UNRELATED) |
| 17 | back | bag (INITIAL C, V) | bike (INITIAL C, FINAL C) | shirt (SEMANTIC) |
| 18 | heart | card (V) | hat (INITIAL C, FINAL C) | nose (SEMANTIC) |
| 19 | seed | keys (V) | sword (INITIAL C, FINAL C) | dirt (SEMANTIC) |
| 20 | book | hood (V) | bike (INITIAL C, FINAL C) | heart (UNRELATED) |
| 21 | duck | bus (V) | dog (INITIAL C, FINAL C) | feet (UNRELATED) |
| 22 | gate | cake (V) | goat (INITIAL C, FINAL C) | bone (UNRELATED) |
| 23 | sword | fork (V) | sad (INITIAL C, FINAL C) | mum (UNRELATED) |
| 24 | card | heart (V) | bird (FINAL C) | leaf (UNRELATED) |
| 25 | road | boat (V) | sword (FINAL C) | shirt (UNRELATED) |
| 26 | pig | bin (V) | leg (FINAL C) | sheep (SEMANTIC) |
| 27 | tongue | bug (V) | king (FINAL C) | chip (SEMANTIC) |
| 28 | dog | rock (V) | dad (INITIAL C) | sheep (SEMANTIC) |
| 29 | pen | leg (V) | pig (INITIAL C) | chalk (SEMANTIC) |
| 30 | leaf | knees (V) | lock (INITIAL C) | man (UNRELATED) |
| 31 | man | sad (V) | mouse (INITIAL C) | shark (UNRELATED) |
| 32 | mum | bus (V) | mouse (INITIAL C) | lock (UNRELATED) |
| 33 | leg | pen (V) | lick (INITIAL C) | shop (UNRELATED) |

| # | Target | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | bug | hug (V, FINAL C) | bag (INITIAL C, FINAL C) | hood (UNRELATED) |
| 2 | goat | boat (V, FINAL C) | gate (INITIAL C, FINAL C) | pig (SEMANTIC) |
| 3 | cat | hat (V, FINAL C) | cot (INITIAL C, FINAL C) | bird (SEMANTIC) |
| 4 | chin | bin (V, FINAL C) | chip (INITIAL C, V) | back (SEMANTIC) |
| 5 | knees | keys (V, FINAL C) | leaf (V) | head (SEMANTIC) |
| 6 | ship | chip (V, FINAL C) | lick (V) | hug (UNRELATED) |
| 7 | hat | bat (V, FINAL C) | house (INITIAL C) | lick (UNRELATED) |
| 8 | hug | bug (V, FINAL C) | hat (INITIAL C) | seed (UNRELATED) |
| 9 | rock | sock (V, FINAL C) | road (INITIAL C) | boot (UNRELATED) |
| 10 | meat | feet (V, FINAL C) | mouse (INITIAL C) | dad (UNRELATED) |
| 11 | light | kite (V, FINAL C) | leg (INITIAL C) | house (SEMANTIC) |
| 12 | dad | sad (V, FINAL C) | dirt (INITIAL C) | mum (SEMANTIC) |
| 13 | chalk | fork (V, FINAL C) | chin (INITIAL C) | pen (SEMANTIC) |
| 14 | mouse | house (V, FINAL C) | man (INITIAL C) | cat (SEMANTIC) |
| 15 | bone | boat (INITIAL C, V) | book (INITIAL C) | tap (UNRELATED) |
| 16 | bag | bat (INITIAL C, V) | hug (FINAL C) | meat (UNRELATED) |
| 17 | boot | bag (INITIAL C) | bat (INITIAL C, FINAL C) | feet (SEMANTIC) |
| 18 | sheep | meat (V) | shop (INITIAL C, FINAL C) | dog (SEMANTIC) |
| 19 | bike | light (V) | book (INITIAL C, FINAL C) | road (SEMANTIC) |
| 20 | lick | ship (V) | lock (INITIAL C, FINAL C) | sword (UNRELATED) |
| 21 | cot | rock (V) | cat (INITIAL C, FINAL C) | bone (UNRELATED) |
| 22 | shop | cot (V) | ship (INITIAL C, FINAL C) | head (UNRELATED) |
| 23 | bus | duck (V) | nurse (FINAL C) | tap (UNRELATED) |
| 24 | hood | book (V) | bird (FINAL C) | knife (UNRELATED) |
| 25 | nurse | bird (V) | bus (FINAL C) | seed (UNRELATED) |
| 26 | nose | coat (V) | keys (FINAL C) | tongue (SEMANTIC) |
| 27 | king | bin (V) | tongue (FINAL C) | sword (SEMANTIC) |
| 28 | knife | light (V) | night (INITIAL C, V) | cake (SEMANTIC) |
| 29 | shark | heart (V) | shop (INITIAL C) | goat (SEMANTIC) |
| 30 | cake | gate (V) | king (INITIAL C) | tongue (SEMANTIC) |
| 31 | tap | back (V) | tongue (INITIAL C) | dog (UNRELATED) |
| 32 | head | leg (V) | heart (INITIAL C) | boot (UNRELATED) |
| 33 | bird | nurse (V) | boot (INITIAL C) | tap (UNRELATED) |

- No included target words fell below 75% correct on receptive testing
- Only *chin* and *ship* exceeded 3.5 s median receptive latency
- Poor spontaneous naming did not necessarily predict poor receptive identification
  - Ten included items had <25% expressive target-word inclusion but still achieved 85% ± 5% receptive accuracy
  - Expressive and receptive latencies showed weak but significant correlation (r = .35, p < .01)

**Table 2:** Foil category distribution in the two finalised word lists

| Category | Set 1 | Set 2 | Total |
|---|---|---|---|
| Initial C only | 15 | 15 | 30 |
| Initial C, V | 4 | 4 | 8 |
| V only | 18 | 18 | 36 |
| Final C only | 5 | 6 | 11 |
| V, final C | 14 | 14 | 28 |
| Initial C, final C | 10 | 9 | 19 |
| Semantic | 15 | 16 | 31 |
| Unrelated | 18 | 17 | 35 |
| **Initial C total** | 29 | 28 | 57 |
| **V total** | 36 | 36 | 72 |
| **Final C total** | 29 | 29 | 58 |

**Table 3:** Vowel category distribution in the two finalised word lists

| Vowel | Set 1 | Set 2 | Total Targets | Targets % |
|---|---|---|---|---|
| TRAP | 4 | 5 | 9 | 13.6% |
| FLEECE | 4 | 3 | 7 | 10.6% |
| KIT | 3 | 4 | 7 | 10.6% |
| LOT | 3 | 3 | 6 | 9.1% |
| STRUT | 3 | 3 | 6 | 9.1% |
| GOAT | 3 | 3 | 6 | 9.1% |
| PRICE | 2 | 3 | 5 | 7.6% |
| NURSE | 2 | 2 | 4 | 6.1% |
| THOUGHT | 2 | 1 | 3 | 4.5% |
| START | 2 | 1 | 3 | 4.5% |
| DRESS | 2 | 1 | 3 | 4.5% |
| MOUTH | 1 | 1 | 2 | 3.0% |
| FOOT | 1 | 1 | 2 | 3.0% |
| FACE | 1 | 1 | 2 | 3.0% |
| GOOSE | 0 | 1 | 1 | 1.5% |
| **Total** | 33 | 33 | 66 | 100% |

## Conclusion

- The UC4AFC provides robust four-alternative monosyllabic stimuli that are more suitable for New Zealand and Australian children than previous North American materials.
- These findings support its use in paediatric speech discrimination assessment and future APD evaluation.

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COI: None declared